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PRACTICAL WEEK 6

WEEK -5 Create a client-server application using Named Pipes (FIFOs). The client sends messages and the server processes and responds to them. Analyze FIFO behavior when multiple clients communicate simultaneously.

Code:

```
UW PICO 5.09 File: 53.c Modified
}
buffer(bytes_read) = '\0';
printf("Client message: %s\n", buffer);
/* Exit command */
if (strcmp(buffer, "exit") == 0)
{
    const char *response = "Server shutting down.";
    write(response_fd, response, strlen(response) + 1);
    break;
}
/* Process the message */
char response[BUFFER_SIZE];
snprintf(response, BUFFER_SIZE,
         "Server processed: %s", buffer);
write(response_fd, response, strlen(response) + 1);
}
close(request_fd);
close(response_fd);
unlink(REQUEST_FIFO);
unlink(RESPONSE_FIFO);
printf("Server terminated.\n");
return 0;
}

Get Help    WriteOut    Read File    Prev Pg    Cut Text    Cut Pos
Exit        Justify     Where is     Next Pg    UnCut Text  To Spell
```

Output:

```
mrk@mallas-mac-Book-Air ~ % nano 53.c
mrk@mallas-mac-Book-Air ~ % gcc 53.c -o 53
mrk@mallas-mac-Book-Air ~ % ./53
Server started.
Waiting for client messages...

```

2. Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

Code:

```
Use PID 5.09 File: 55.c Modified
printf("Waiting for signals...\n");
printf("Press Ctrl+C to generate SIGINT.\n");
while (1)
{
    printf("Program is running...\n");
    sleep(3);
}
return 0;
}
```

Output:

```
Last login: Tue Sep 22 11:56:42 on ttys000
/nrk@nallas-mac-Book-Air ~ % nano 55.c

/nrk@nallas-mac-Book-Air ~ % gcc 55.c -o 55
/nrk@nallas-mac-Book-Air ~ % ./55
POSIX Signal Handling Program
Process ID (PID): 10420
Waiting for signals...
Press Ctrl+C to generate SIGINT.
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
```

